



Final Review

Today's Agenda

- Questions/Discussion on Assignment 2
- Review DQN (with num_blocks X num_positions block-stacking)
- Review RL Slides
- About Written Questions

RL Diagrams with virtual Create 3

Loaner Laptop
ROS 2 Nodes Communicating

action A_t

Agent:
Q-Learning or
Stable-baselines3:
DQN or PPO or...

R_t
 S_t

R_{t+1}
 S_{t+1}

Gymnasium Environment

Step:
publish to cmd_vel
Wait for stop_status
Process image_raw
Return observation + reward

ROS 2
Node

Gazebo
AWS Small House

Camera
image_raw

Create
3

DQN (num_blocks X num_positions)

DQN with large state space (6 blocks, 6 positions):

- Limit episodes to 200 steps
 - In early training, agent sees more block configurations
- learning_starts=2000
 - Perform 2000 steps before training Q-network

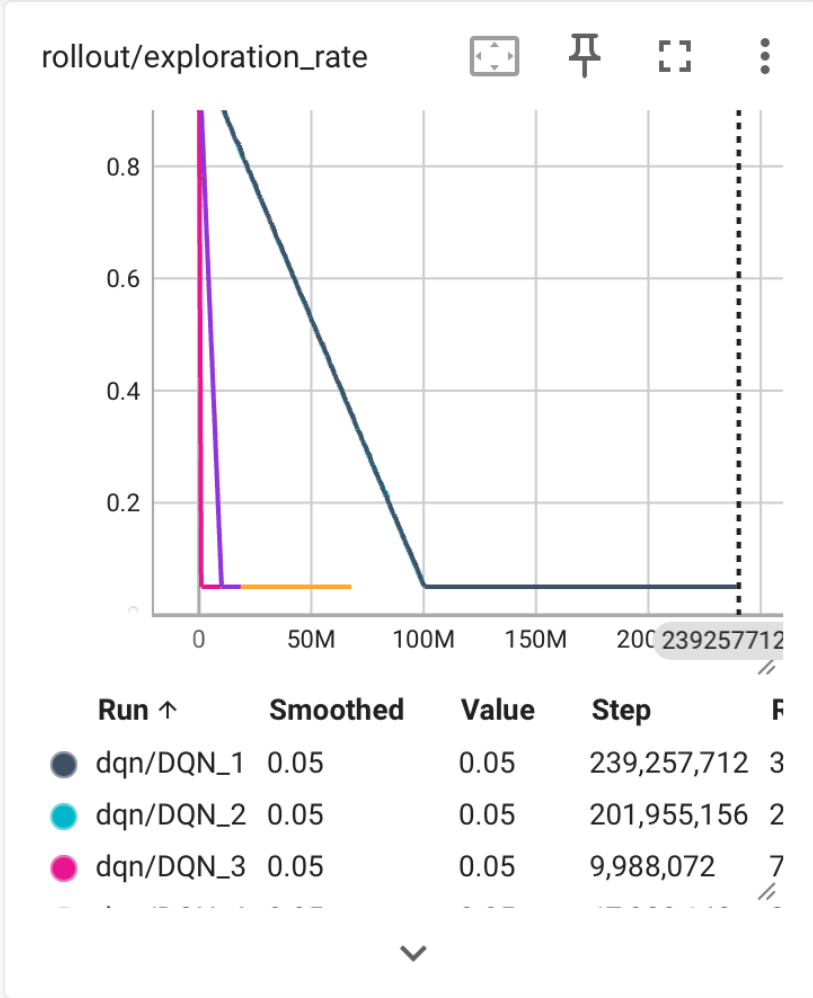
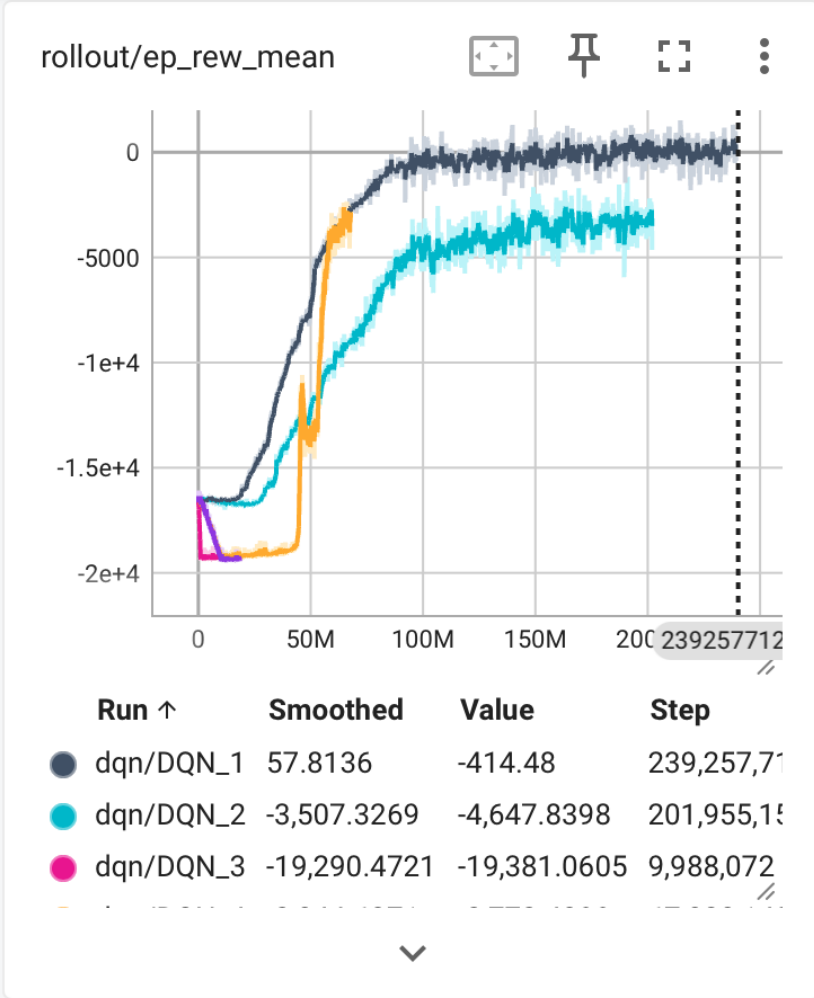
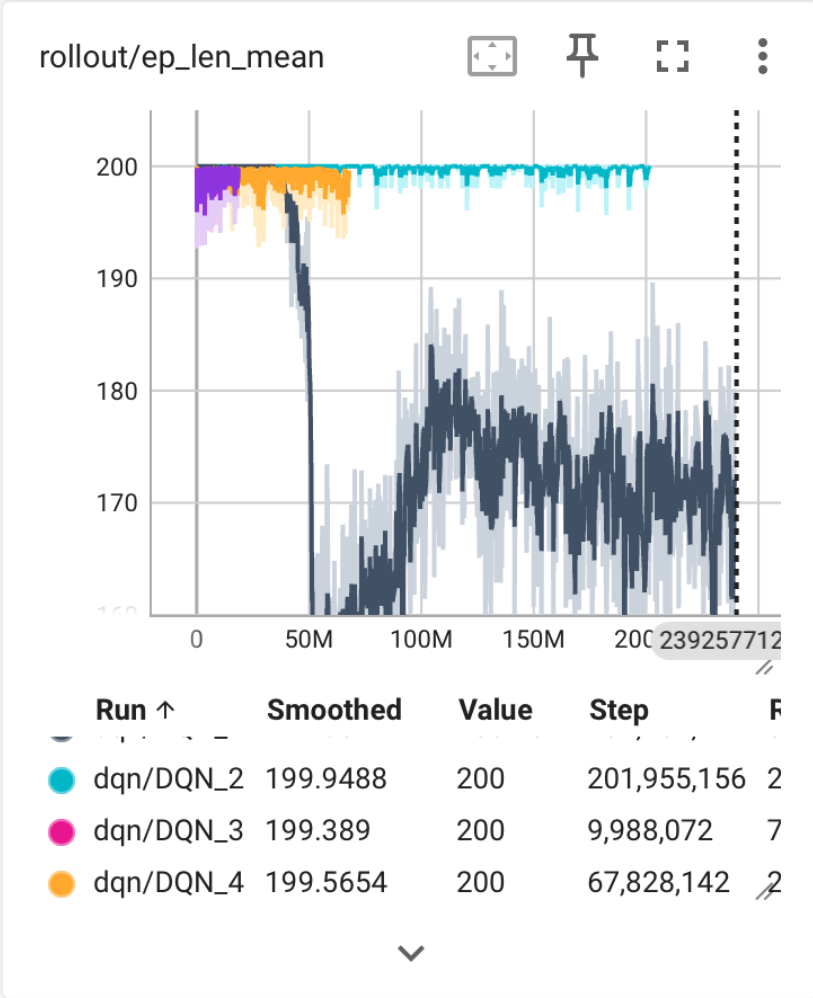
Filter runs (regex)

Filter tags (regex)

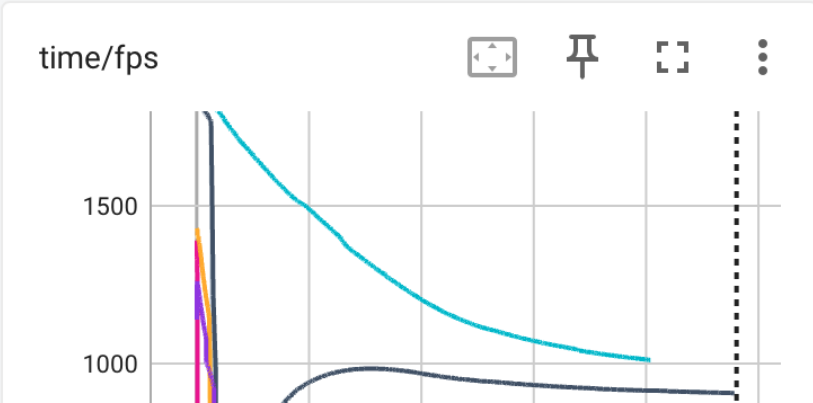
All

- ☒ Run ↑
- ☒ dqn/DQN_1
- ☒ dqn/DQN_2
- ☒ dqn/DQN_3
- ☒ dqn/DQN_4
- ☒ dqn/DQN_5

rollout 3 cards



time



DQN on 6x6 block-stacking

- Curriculum Learning: Make problem easier at the beginning and increase difficulty in subsequent additional training runs:
 - Keep same dimensionality 6x6 (needs to be same environment, just easier)
 - Introduce “difficulty level” : number of steps between start and target (DQN_4 run has difficulty_level=2)
 - Introduce “start_flat” : mode where every episode starts with block on its corresponding position, forcing num_positions=num_blocks (DQN_4 run has start_flat=True)

DQN Q-network structure

- Default structure is [64,64]
- Q-network structure is increased to [512,512,256] in DQN5 run
 - Current configuration: 6 1-hot vectors of length 12
 - Target configuration: 6 1-hot vectors of length 12
 - MultiInputPolicy -> input is vector of length 144
- Capacity: the first 512-layer has enough surface area to learn specific combinations of block positions
- Separation: Deep layers (3 total) help the model learn that a 1 in the 7th bit (Block a on Position 1) combined with a 1 in the 13th bit (Block b on Block a) represents a stack

Slides

CST8509_01_RL_Intro

- Basic concepts and definitions

CST8509_02_MDP

- MDPs, more on Value functions, Policy

CST8509_03_Gymnasium

- Practical Agent – Environment infrastructure, Prolog, Pygame

CST8509_04_StableBaselines3

- SB3 features: Wrappers, Callbacks, Vectorized Environments

Slides (cont'd)

CST8509_05_DQN_StableBaselines3

- DQN practical aspects

CST8509_07_Gazebo_DynamicP_MC

- Gazebo Simulation, Dynamic Programming, Monte Carlo Methods

CST8509_08_Docker

- Solution for python virtual environments and ROS2 workspaces

CST8509_09_Value_Fn_approx

- Realistic large State/Action spaces
- DQN and value function approximation
- PPO and value function approximation

About Written Questions

- Example Written Questions from Midterm and Midterm Review
- "Time to check your learning" slides
- Some Multiple Choice questions can be rephrased into written questions

Written Question Additional Examples

How can Reinforcement Learning be applied to problems where the number of states is huge?

Describe at a high level how the DQN algorithm does value function approximation

Describe at a high level how the PPO algorithm does value function approximation

Where would a neural network appear in a Reinforcement Learning Problem/Solution?

What are possible sources of training data for a neural network in a Reinforcement Learning Problem/Solution?

Written Question Additional Examples

- Give an overview of the strategy used in Assignment 2 to set up a Reinforcement Learning solution with simulation for training a Create3 robot to do some arbitrary task, for example, the red-ball-following task of Assignment 2. Your overview should mention the major tools, technologies, and projects used, their purpose and how they interact in the solution.
- Suppose you are running an SB3 agent on a gymnasium environment with unsatisfactory results. What areas could you explore in order to improve results?